

Literature-Implied Partial Subcase for arXiv:1701.07084

FA/Banach run fa.banach_001

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Status

This is a `literature_implied_answer (partial subcase)` packet. It does not claim an original proof or a full answer to Correa's final question for arbitrary Banach-space interpolation pairs. It records that a later splitting-stability theorem of Castillo–Correa–Ferenczi–Gonzalez [2] gives an affirmative answer for interpolation pairs of superreflexive Kothe function spaces.

Source Question

In Section 5, Final Remarks, Correa [1] asks:

If dX_θ is nontrivial for every $\theta \in (0, 1)$, $\theta \neq \frac{1}{2}$, is $dX_{1/2}$ nontrivial?

In the source paper, dX_θ denotes the derived twisted sum associated with the complex interpolation derivation Ω_θ .

Supporting Theorem

The decisive supporting result is Theorem 4.3 of [2], on PDF page 16: if (X_0, X_1) is an interpolation pair of superreflexive Kothe function spaces and $0 < \theta < 1$, then Ω_θ is trivial if and only if there is a weight function w such that

$$X_1 = X_0(w)$$

up to equivalence of norms.

Identification

The endpoint condition $X_1 = X_0(w)$ is independent of θ . Therefore Theorem 4.3 has the following immediate contrapositive consequence:

If (X_0, X_1) is a superreflexive Kothe interpolation pair and dX_θ is nontrivial for at least one $0 < \theta < 1$, then the endpoint pair is not merely a weighted version of one Kothe space; hence no Ω_t , $0 < t < 1$, is trivial.

In particular, if dX_θ is nontrivial for every $\theta \neq 1/2$, then $dX_{1/2}$ is nontrivial.

Scope and Limitations

This answers the source question only for interpolation pairs of superreflexive Kothe function spaces. That includes the standard ℓ_p - and L_p -type Kothe settings motivating the question, but it does not settle the arbitrary Banach-space pair case. The supporting paper explicitly notes that the existence of local or global stability for the differential process associated to complex interpolation of a general couple of Banach spaces remains open.

The relation to Correa’s final question is an agent-identified implication. The supporting paper cites Correa’s article and develops the relevant splitting-stability theory, but it does not single out the final midpoint question as the named problem being answered.

Search Bounds

Before packaging, the run’s cheap indexes were searched for 1701.07084, the paper title, $dX_{\{1/2\}}$, **nontrivial for every theta**, and core twisted-sum/interpolation keywords. Local later-source searches for citations to Correa’s paper found arXiv:1712.09647 as the decisive supporting theorem. No prior packet or attempt for this exact Kothe subcase was found.

Human Review Recommendation

Review as a provenance/status packet. The main points to check are the translation between dX_θ and the derivation-generated exact sequence for Ω_θ , and that the packet is restricted to superreflexive Kothe function-space interpolation pairs.

References

- [1] W. H. G. Correa, *Type, cotype and twisted sums induced by complex interpolation*, arXiv:1701.07084.
- [2] J. M. F. Castillo, W. H. G. Correa, V. Ferenczi, and M. Gonzalez, *Stability of the differential process generated by complex interpolation*, arXiv:1712.09647.